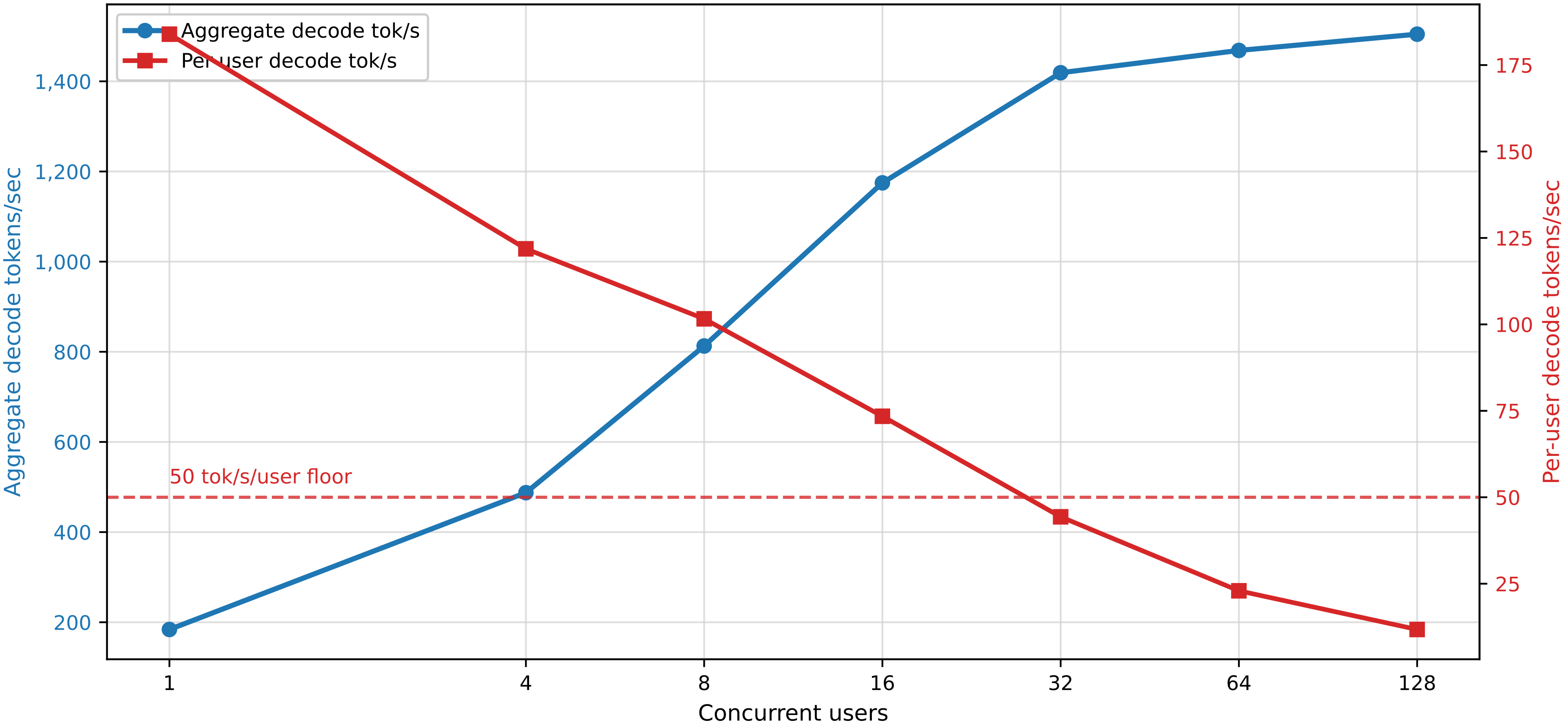
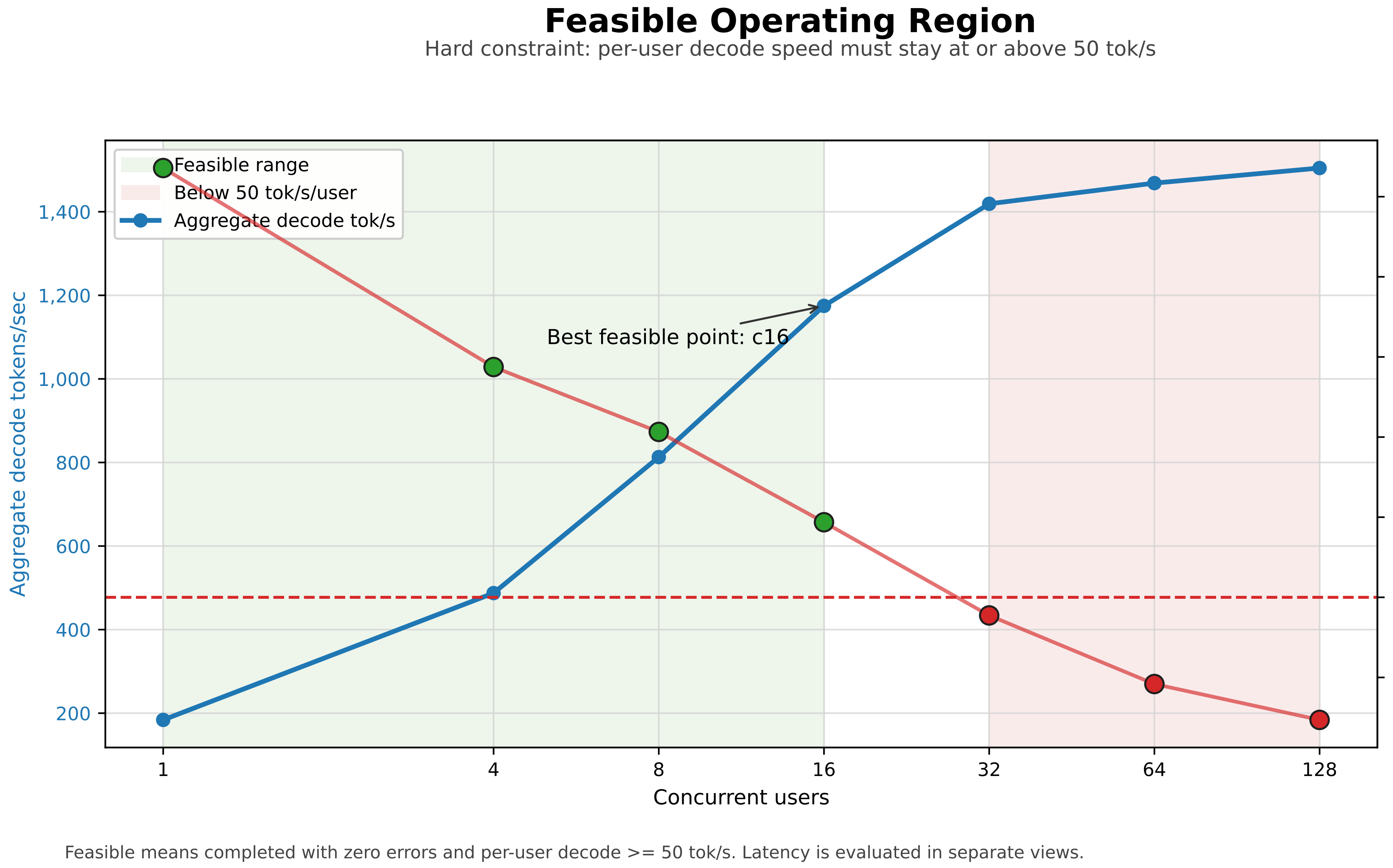
Dual-Axis Throughput Tradeoff

64k active decode: aggregate throughput rises as per-user speed falls



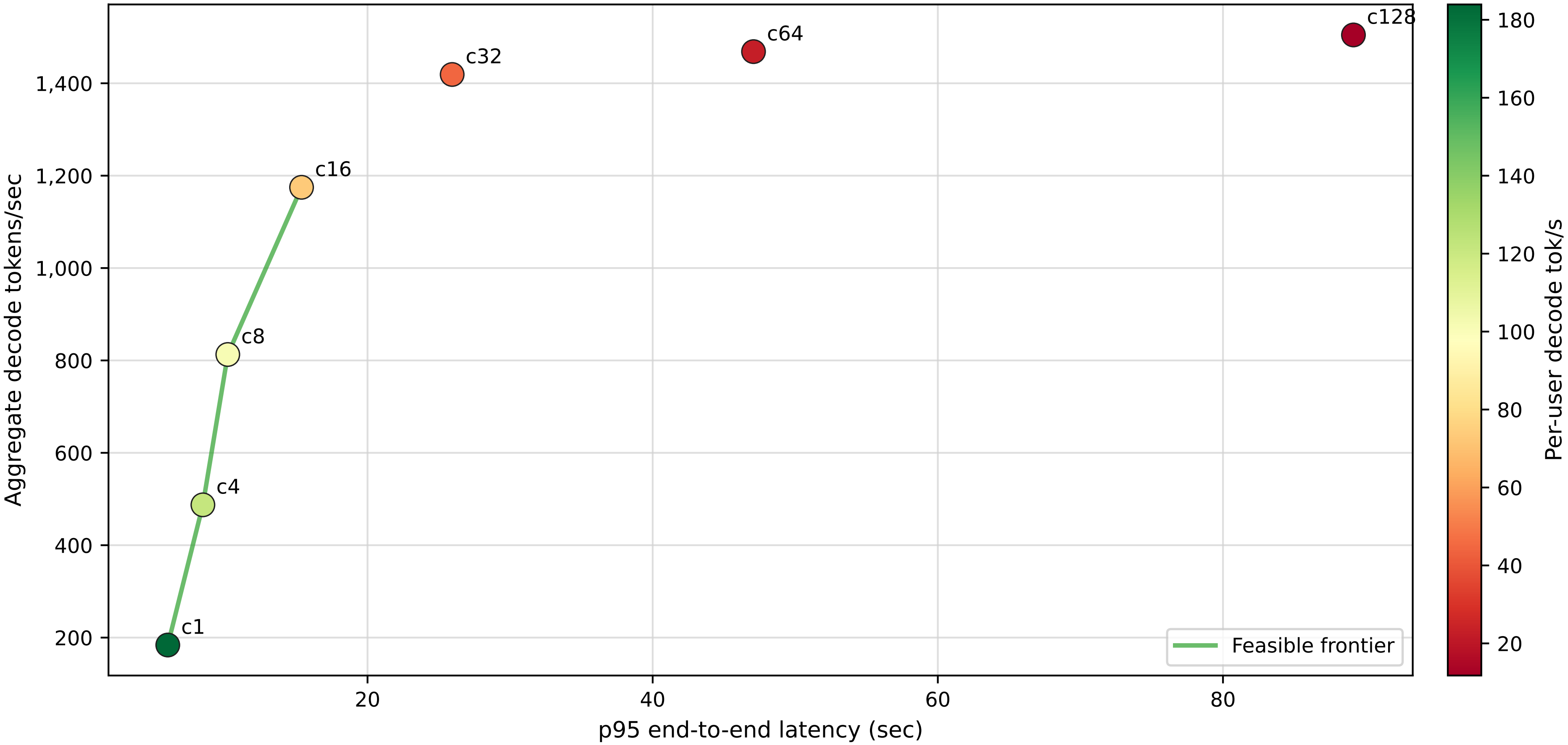
Source: localperf qwen36-35b-nvfp4-hf-endpoint.sqlite. Selected 64k rows use completed zero-error measurements.



Feasible Operating Region
Hard constraint: per-user decode speed must stay at or above 50 tok/s
Feasible range
Below 50 tok/s/user
Aggregate decode tok/s
Best feasible point: c16
Aggregate decode tokens/sec
Per-user decode tokens/sec
Concurrent users
1
4
8
16
32
64
128
200
400
600
800
1,000
1,200
1,400
25
50
75
100
125
150
175
Feasible means completed with zero errors and per-user decode >= 50 tok/s. Latency is evaluated in separate views.

Pareto View: Throughput vs Latency

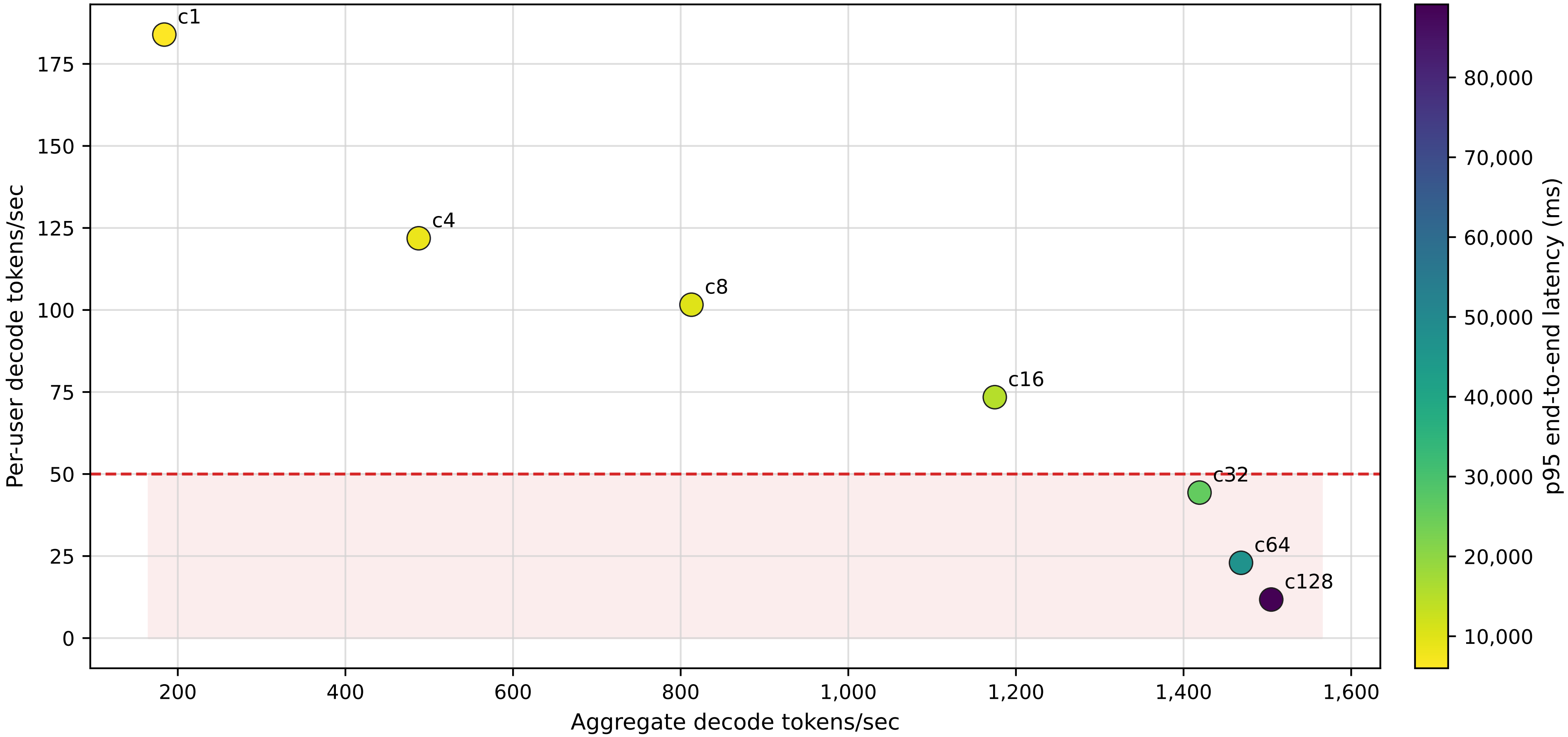
Higher is better on Y; farther right is worse latency



A point is operationally useful only if it satisfies the per-user floor and any latency SLO you choose.

Aggregate vs Per-User Tradeoff

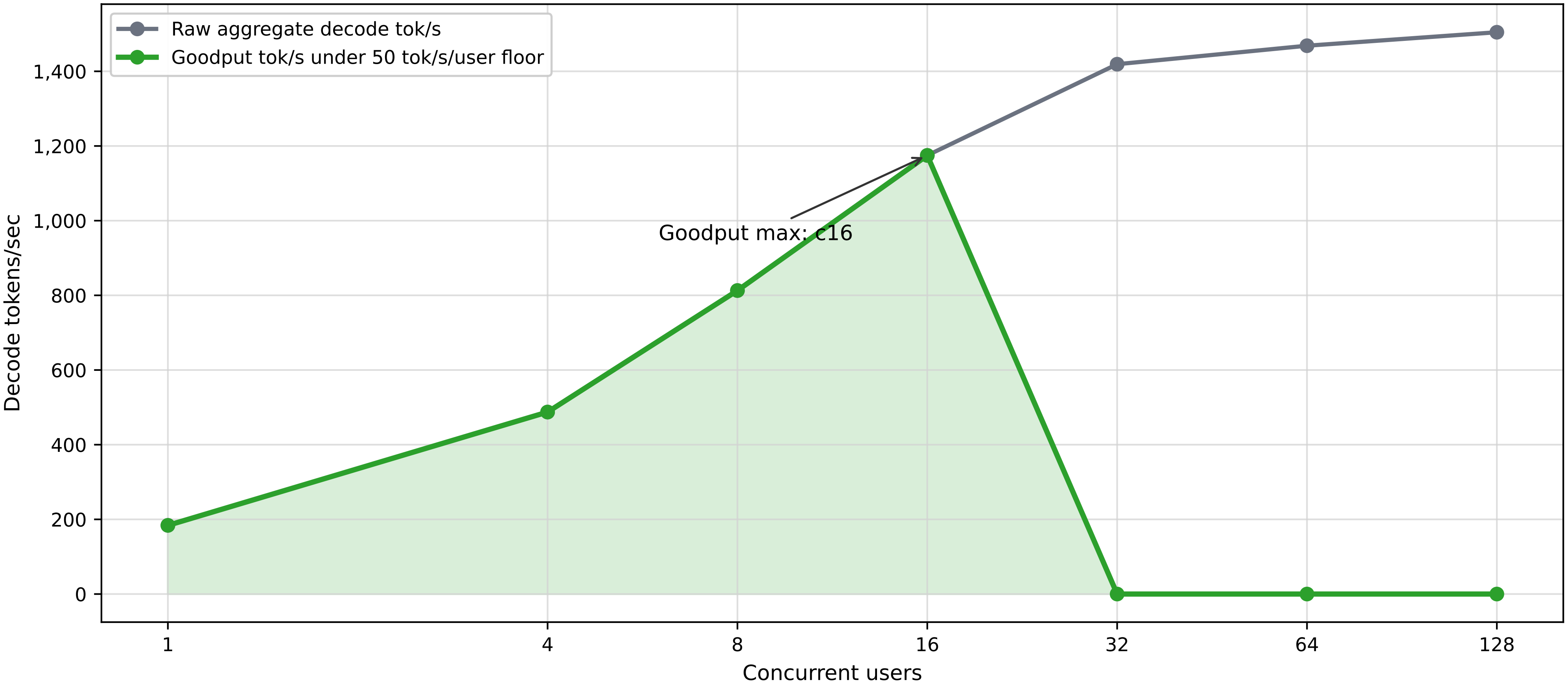
Color shows p95 end-to-end latency; horizontal line is the 50 tok/s/user floor



This is the clearest single chart for explaining why c32+ are high-throughput but below the user-speed floor.

Goodput Curve

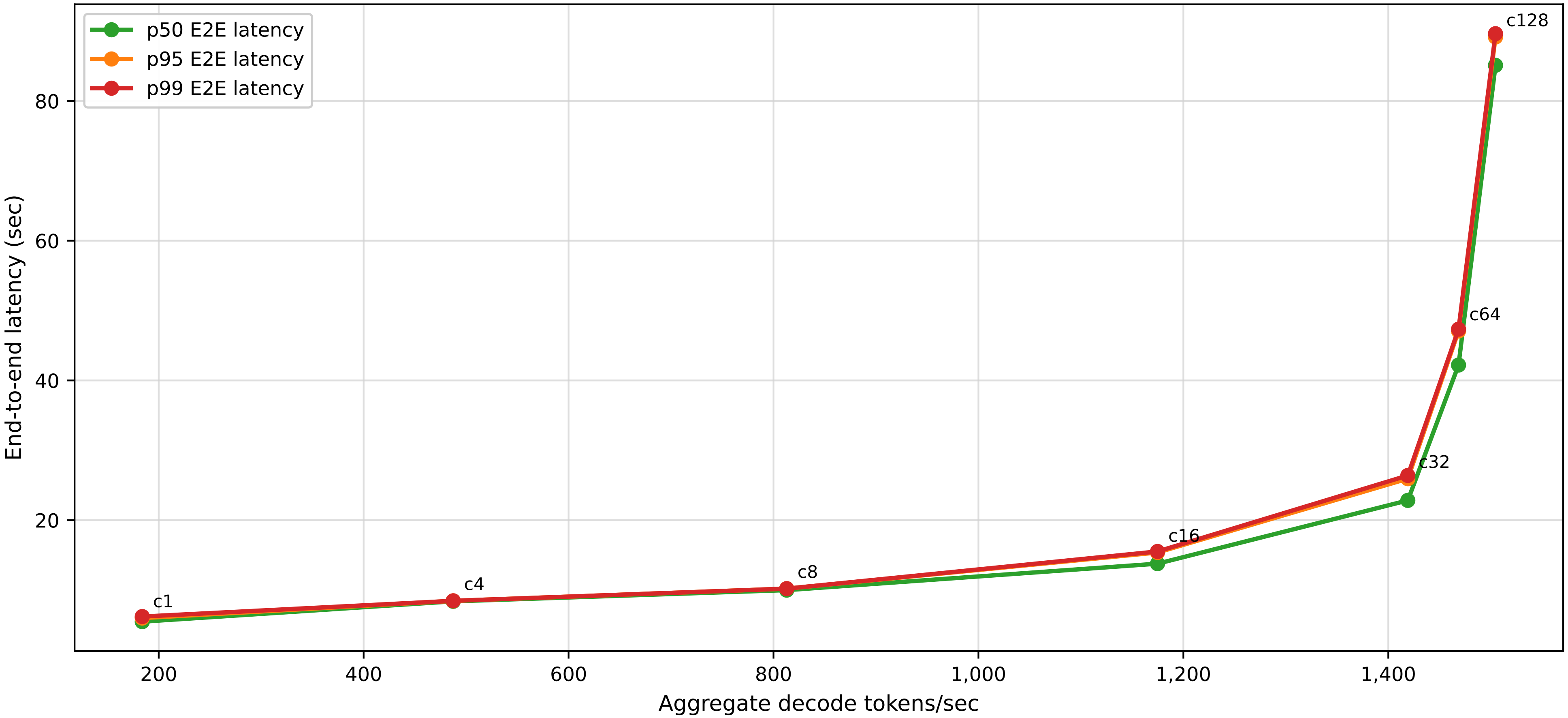
Goodput counts aggregate tokens only while per-user speed is at least 50 tok/s and errors are zero



If you add a p95 latency SLO, this same curve should zero out points that violate that SLO too.

Latency-Throughput Curve

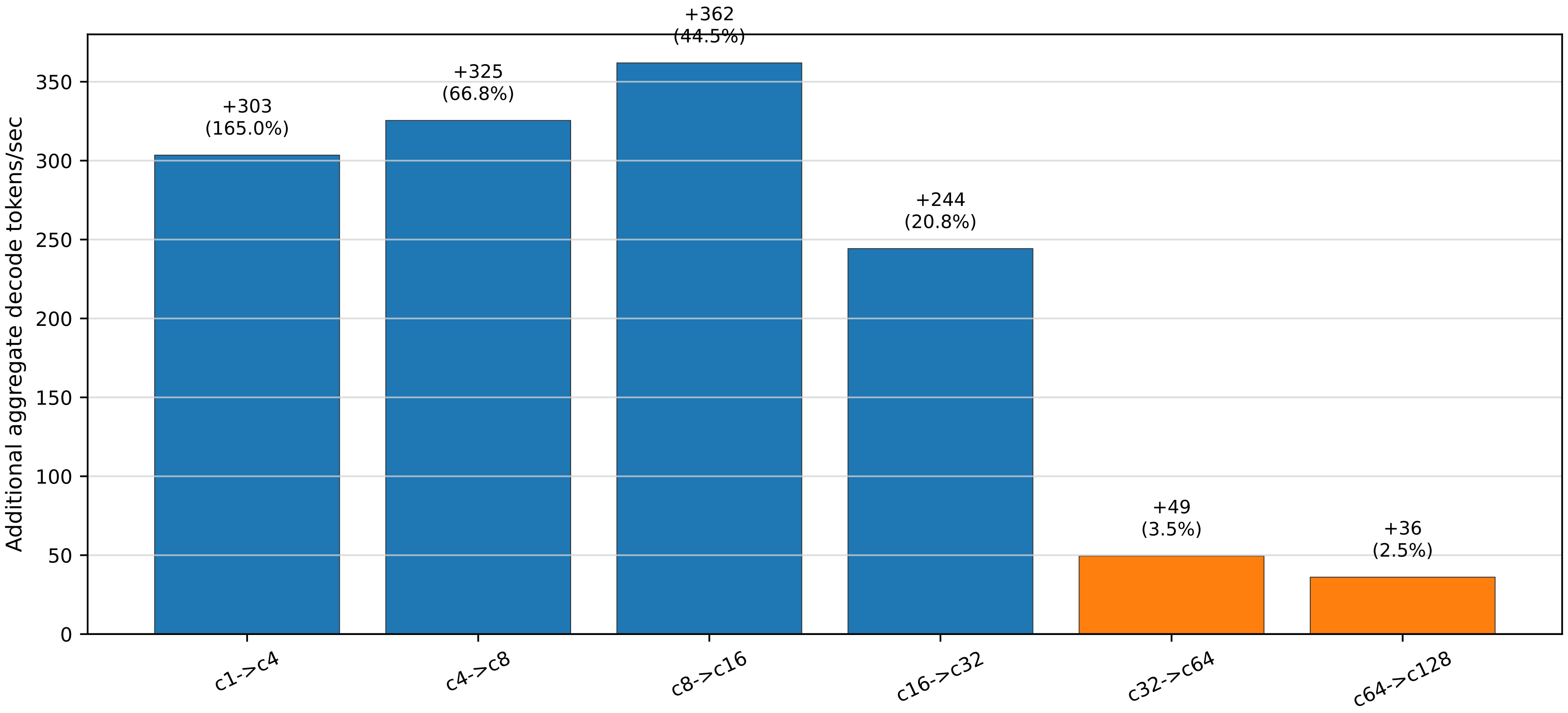
Shows where latency bends upward as aggregate decode throughput increases



This view is useful for setting a latency SLO after choosing the minimum per-user streaming speed.

Knee Plot: Marginal Throughput Gain

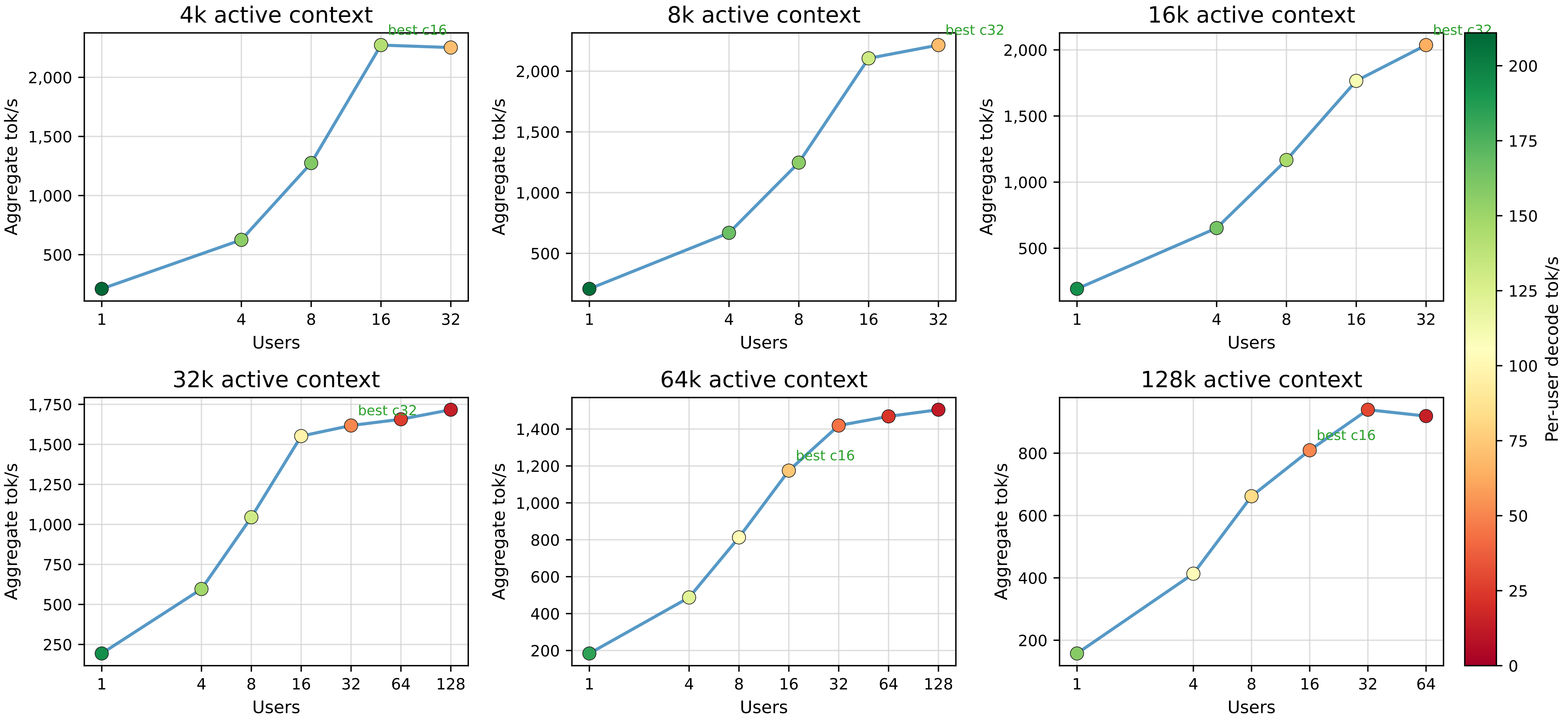
After c32, extra concurrency mostly adds queueing instead of useful throughput



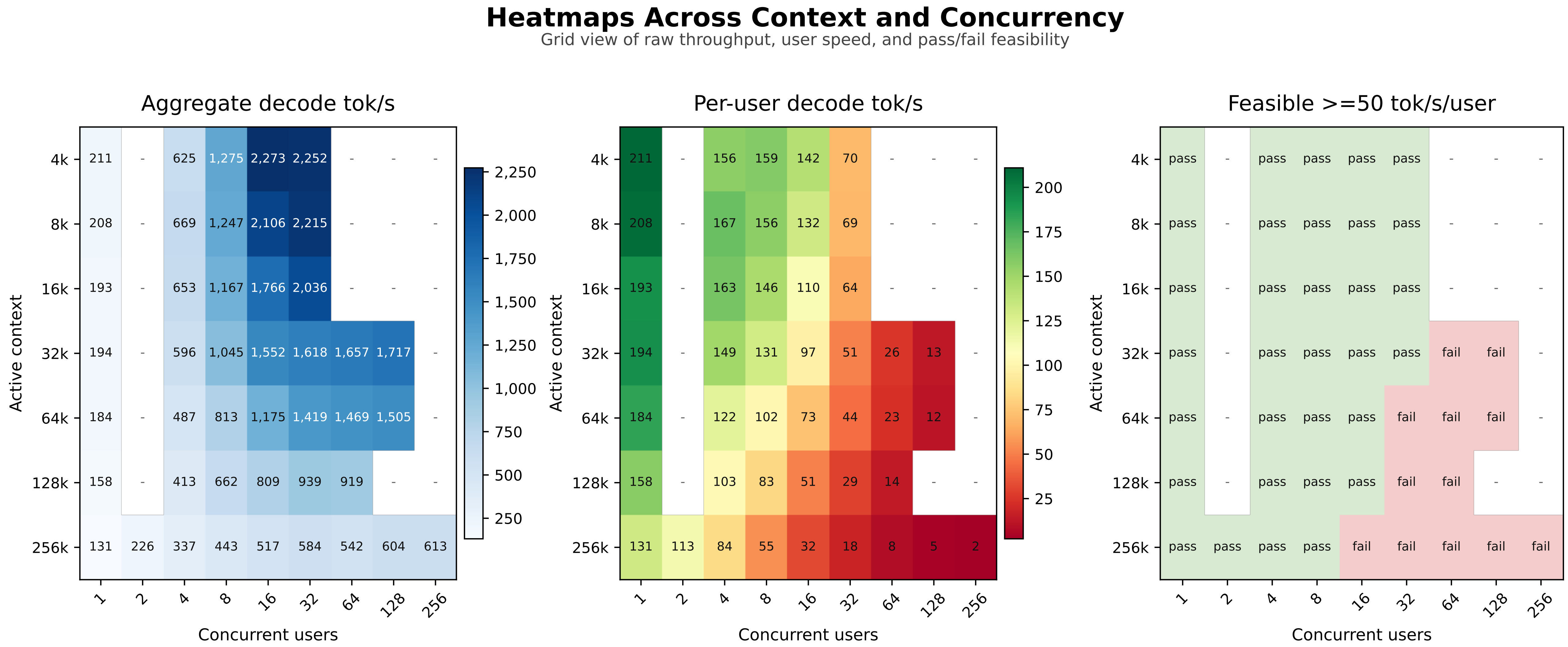
This is the fastest way to see diminishing returns. c64 and c128 add little throughput but greatly increase latency.

Small Multiples by Active Context

Aggregate decode lines with per-user speed encoded by marker color



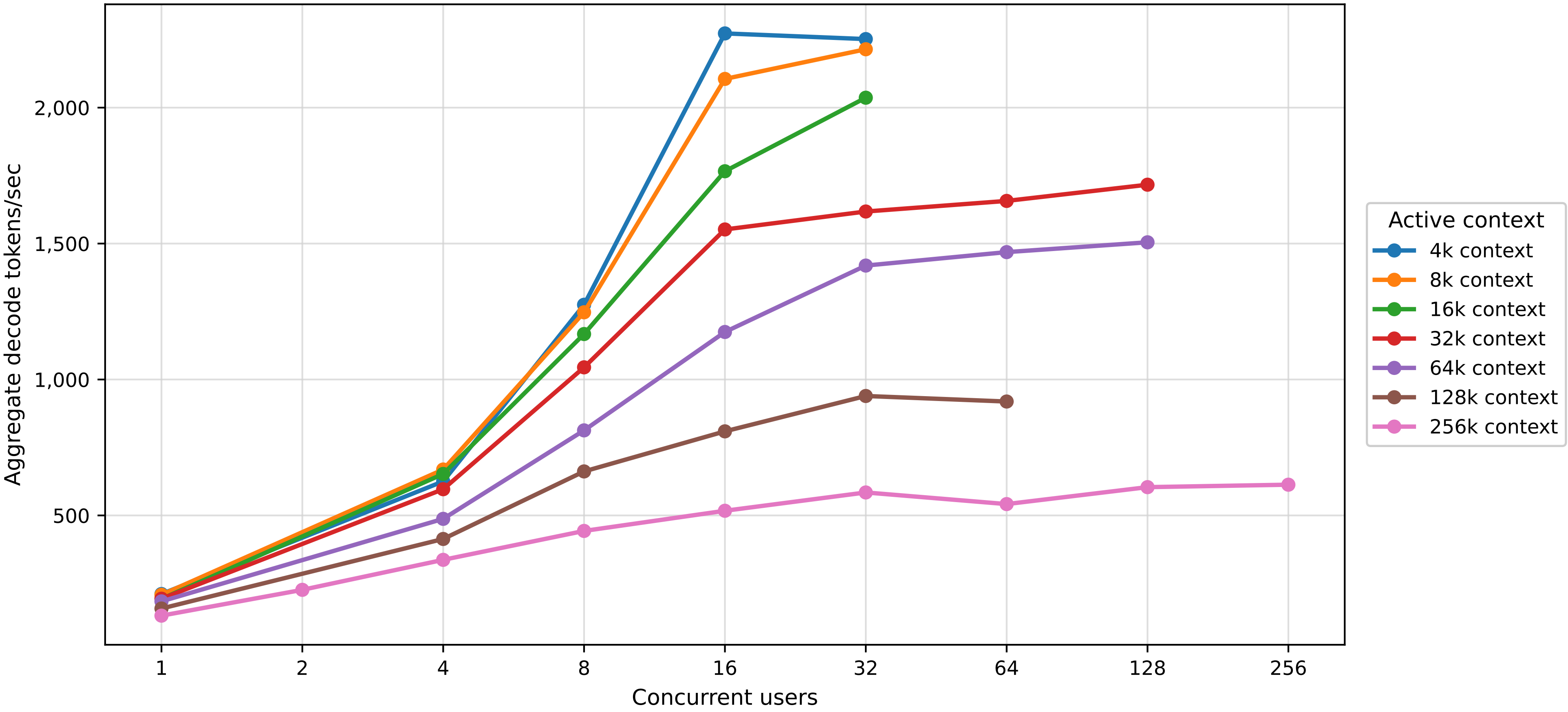
Rows across 128k and 256k include observed measurements from multiple max-num-seqs server profiles; duplicate context/concurrency rows keep the highest aggregate result.



Blank cells were not measured. Feasible only applies the 50 tok/s/user and zero-error rule; add latency SLO if needed.

Aggregate Decode Throughput by Active Context

X-axis is concurrency; each line is one active context size



Duplicate context/concurrency rows keep the highest completed zero-error aggregate result. 128k/256k include measurements from multiple server profiles.